

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2024

Programme: BE

Full Marks : 100

Course: Image Processing and Pattern Recognition

Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define digital image processing? Explain briefly about Components of an Image Processing System. 8
- b) What is Neighborhood Operations in Image Enhancement? Derive Laplacian Filter Mask by using second order partial differential equation. 7
2. a) What is zooming? Illustrate with an example by interpolation and replication. 8
- b) What is histograms equalization? Perform the histogram equalization on the following 8-bit gray level image and also scale the intensity from 1 to 20. 7

3	2	4	5
1	2	1	2
7	3	1	2
7	6	4	7

3. a) What are the steps involved in frequency domain filtering? Explain the essential properties of 2-D Discrete Fourier Transform. 8
- b) What is Image Restoration? Explain any two Noise models along with their equations. 7
4. a) From the following probability distribution construct the Huffman Code for each gray level and find efficiency of the coding. 8

r_k	0	1	2	3	4	5
$p(r_k)$	0.05	0.1	0.25	0.15	0.3	0.15

- b) What do you mean by lossless image compression? Briefly explain predictive coding. 7

5. a) Define structuring elements, dilation and erosion in morphological image processing. Explain opening and closing with suitable examples. 7
- b) What do you mean by image segmentation, give the algorithm for basic adaptive thresholding. 8
6. a) What are two approaches of image segmentation? Explain the process of image segmentation by region splitting and merging. 8
- b) Define pattern and pattern recognition system; explain the issues with feature selection. 7
7. Write short notes on: (**Any two**) 2×5
- a) Template Matching
- b) Steps of Pattern recognition system
- c) Bayesian Classifier